

QUESTION BANK			
SUBJECT: 24MCAOST203 –Advanced operating systems			
SEM/YEAR: II/I			
Module I			
Overview: Functions of Operating System –Design Approaches –Types of Advanced Operating Systems. Synchronization Mechanisms: Concept of Processes and Threads –The Critical Section Problem- Other Synchronization Problems: – Monitor –Serializer – Path Expressions. Distributed Operating Systems: - Issues in Distributed Operating System –Communication Networks And Primitives –Lamport’s Logical clocks – Causal Ordering of Messages.			
Q No.	Questions	BTL	Competence
1	Classify Advanced operating systems.	1	Remembering
2	Describe Lamports logical clocks with a simple example .	1	Remembering
3	Explain the concept of a distributed operating systems and their features,advantages and issues.	1	Remembering
4	Write a note on the following a. mutex b. semaphore	1	Remembering
5	Explain about Lamport’s logical clocks and casual ordering of messages	1	Remembering
6	What is process? Discuss about different states of a process with diagram	1	Remembering
7	What is the concept of path expressions? Explain the purpose of the following path expressions: i) Path read + write end ii) Path write;{read}end	1	Remembering
8	Differentiate between distributed operating system and real time operating systems?	1	Remembering
9	Explain different states of a process with Diagram	1	Remembering
10	Explain about the major features of access control list	1	Remembering
11	Explain Types of Advanced operating system	1	Remembering
12	Explain monitors,serialisers and path expressions in synchronizations.	1	Remembering
13	Explain functions of advanced operating system	1	Remembering
14	What is meant by distributed operating systems? Explain in details any three issues in distributed operating systems.	1	Remembering

Module II

Distributed Mutual Exclusion: Classification - Requirements – Measuring Performance –Lamport’s Algorithm – Rickart-Agarwala Algorithm – SuzukiKasami’s Broadcast Algorithm;Security: Potential Security Violations – Design Principles for Secure Systems –The Access Matrix Model and Implementation- The Access Control list Method.

Q No.	Questions	BTL	Competence
1	Explain about the major features of access control list	2	Understanding
2	Explain some of the algorithms for mutual exclusion.	2	Understanding
3	Compare the Lamport’s algorithm and Rickart-Agarwala algorithm.	2	Understanding
4	Explain any six Design Principles for Secure Systems.	2	Understanding
5	Write short notes on a. Rickart-Agarwala Algorithm b. Lamport’s algorithm.	2	Understanding
6	Describe classification of mutual exclusion algorithm	2	Understanding
7	Discuss about Distributed Mutual Exclusion Classification requirements and measuring Performance	2	Understanding
8	Explain Lamport’s Algorithm for Mutual Exclusion in Distributed System	2	Understanding
9	Compare token and non-Token based algorithm	2	Understanding
10	Describe Suzuki–Kasami Algorithm for Mutual Exclusion in Distributed System	2	Understanding
11	Discuss Access Matrix Model with its access control list method implementation	2	Understanding

Module III

Distributed Resource Management: Mechanisms for building Distributed File Systems –Design Issues - Distributed Shared Memory – Algorithms for Implementing Distributed Shared memory – Issues in Load Distributing – Components of Load Distributing Algorithm– Sender-Initiated Algorithm – Receiver-Initiated Algorithm.

Q No.	Questions	BTL	Competence
1	Explain major design issues and building mechanisms in Distributed file systems.	2	Understanding
2	Explain important algorithms for implementing DSM	2	Understanding
3	Explain issues in load distribution	2	Understanding
4	Write any two algorithms for implementing DSM	2	Understanding
5	Identify major design issues in Distributed File systems.	2	Understanding
6	Compare between load balancing and load sharing	2	Understanding
7	Summarize Distributed Resource Management	2	Understanding
8	With a clear flowchart, explain the receiver-initiated load distribution algorithm in distributed systems	2	Understanding
9	Describe the Mechanisms for building Distributed File Systems	2	Understanding
10	Classify the components of a load distributing algorithm.	2	Understanding
11	Discuss the different algorithms for implementing distributed shared memory	2	Understanding
12	Discuss how to implement distributed shared memory using the read replication algorithm.	2	Understanding
13	Compare read replication and full replication algorithm	2	Understanding
14	Classify Sender Initiated Algorithm and Receiver Initiated Algorithm	2	Understanding
15	Discuss how to implement distributed shared memory using the read replication algorithm.	2	Understanding
16	Explain architecture of distributed file system	2	Understanding
17	Compare mounting and caching of distributed file system	2	Understanding
18	Summarize four algorithms of distributed shared memory	2	Understanding
19	Explain Distributed Scheduling Issues in Load Distributing	2	Understanding
20	Describe various issues of load distributing	2	Understanding
21	Classify load distributing algorithms	2	Understanding
22	Explain Components of Load Distributing Algorithms	2	Understanding
23	Explain design issues such as naming,caching,and replication in distributes file systems.	2	Understanding

Module IV

Multiprocessor Operating Systems: Basic Multiprocessor System Architectures –Interconnection Networks – Structures – Design Issues – Threads – Process - Synchronization– Processor Scheduling – Memory Management – Virtualization – Types of Hypervisors –Para virtualization – Memory Virtualization – I/O Virtualization.

Q No.	Questions	BTL	Competence
1	Explain system architecture of Multiprocessor systems.	2	Understanding
2	Explain design issues in Database Multiprocessor Systems.	2	Understanding
3	Explain how virtualization is implemented.	2	Understanding
4	Explain Multiprocessor System Architectures and Interconnection Networks	2	Understanding
5	Discuss the synchronization of processes in Multiprocessors	2	Understanding
6	Illustrate Basic Multiprocessor System Architectures	2	Understanding
7	Describe Interconnection Networks For Multiprocessor systems	2	Understanding
8	Explain three basic types of multiprocessor operating system	2	Understanding
9	Explain Design issues of operating system	2	Understanding
10	Summarize threads and its types	2	Understanding
11	Compare thread and process	2	Understanding
12	Explain virtualization	2	Understanding
13	Describe Virtualization in Operating Systems and different type of hypervisors	2	Understanding
14	Distinguish the differences between UMA and NUMA architecture of multiprocessor systems.	2	Understanding
15	Discuss about Multiprocessor System Architectures and Interconnection Networks	2	Understanding
16	Write short notes on: I. Memory Virtualization II. Para Virtualization	2	Understanding
17	Explain Hypervisor and their working	2	Understanding

Module V

Database Systems: Problem of Concurrency Control – Serializability – Basic Synchronization Primitives for Concurrency Control – Lock-Based Algorithms – Time Stamp Based Algorithms – Optimistic Algorithms.

Q No.	Questions	BTL	Competence
1	Explain Lock based algorithms for concurrency control in Database Systems	2	Understanding
2	Illustrate Timestamp based algorithms for concurrency control in Database Systems.	2	Understanding
4	Explain the basic Synchronization Primitives for Concurrency Control in Database systems.	2	Understanding
5	Write and explain an optimistic algorithm for concurrency control in database systems.	2	Understanding
6	Explain the basic synchronization primitives for concurrency control algorithms in database systems?	2	Understanding
7	Compare Lock based algorithm and Time Stamp based algorithms	2	Understanding
8	Discuss the synchronization of processes in Multiprocessors	2	Understanding
9	What is Serializability and explain Basic Synchronization Primitives for Concurrency Control	2	Understanding
10	Illustrate concurrency control algorithms.	2	Understanding
11	Discuss the main issues in multiprocessor operating system	2	Understanding
12	Explain the Kung-Robinson concurrency control algorithm	2	Understanding
13	Explain the core principles of optimistic concurrency control algorithm	2	Understanding